

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.035508089s cost=1.41429382399 lag_gradient={'rotor':
-0.3029846273480847, 'gimbal': -2.634008490265405e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.387778780840665e-17, 'analytic_l2_norm':
7.9717198621302385, 'finite_difference_l2_norm': 0.0, 'difference_l2_norm':
7.9717198621302385, 'relative_error': 1.0}, {'direction': 'rotor_delay_seconds', 'scheme':
'forward', 'step': 1.3877787807814457e-17, 'analytic_l2_norm': 7.971731435020553,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 7.971731435020553, 'relative_error':
1.0}, {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step':
6.0554544523933395e-06, 'analytic_l2_norm': 17.20508236588638, 'finite_difference_l2_norm':
17.2050822851243, 'difference_l2_norm': 6.97912807359031e-07, 'relative_error':
4.056433979896705e-08}]

smooth width=2: rotor=0.2s gimbal=0.035373404s cost=1.41562078066 lag_gradient={'rotor':
-0.30943002181043744, 'gimbal': 3.819073831845543e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.387778780890199e-17, 'analytic_l2_norm':
8.764019391342663, 'finite_difference_l2_norm': 0.0, 'difference_l2_norm':
8.764019391342663, 'relative_error': 1.0}, {'direction': 'rotor_delay_seconds', 'scheme':
'forward', 'step': 1.3877787807814457e-17, 'analytic_l2_norm': 8.764006959236921,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 8.764006959236921, 'relative_error':
1.0}, {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step':
6.0554544523933395e-06, 'analytic_l2_norm': 19.010281380757547, 'finite_difference_l2_norm':
19.010280624844683, 'difference_l2_norm': 5.2255057429364035e-06, 'relative_error':
2.748778746760544e-07}]

smooth width=1: rotor=0.2s gimbal=0.0354760315s cost=1.41586228947 lag_gradient={'rotor':
-0.13975287011138368, 'gimbal': -5.844775814070613e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.3877787819984907e-17,
'analytic_l2_norm': 10.909099655638693, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 10.909099655638693, 'relative_error': 1.0}, {'direction':
'rotor_delay_seconds', 'scheme': 'forward', 'step': 1.3877787807814457e-17,
'analytic_l2_norm': 10.909140071717017, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 10.909140071717017, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 26.437388537955943, 'finite_difference_l2_norm': 26.437369126555442,
'difference_l2_norm': 2.6615628304924346e-05, 'relative_error': 1.0067419581443344e-06}]

smooth width=0.5: rotor=0.2s gimbal=0.0352183658s cost=1.41584366376 lag_gradient={'rotor':
-0.06101303447052212, 'gimbal': -0.015443049343798407} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.43154600402816e-17, 'analytic_l2_norm':
16.77401079875116, 'finite_difference_l2_norm': 0.0, 'difference_l2_norm':
16.77401079875116, 'relative_error': 1.0}, {'direction': 'rotor_delay_seconds', 'scheme':
'forward', 'step': 1.3877787807814457e-17, 'analytic_l2_norm': 14.678103118837408,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 14.678103118837408,
'relative_error': 1.0}, {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step':
6.0554544523933395e-06, 'analytic_l2_norm': 38.59360103505382, 'finite_difference_l2_norm':
38.609740170494916, 'difference_l2_norm': 1.1199114519178093, 'relative_error':
0.02900593080845521}]

strict profile iterations: 1

iteration 1: best=(0.2, 0.040217049712) cost=1.4156315077 rotor_range=(0.195002269745, 0.2)
gimbal_range=(0.030219681854, 0.045215733641) expansions=[{'best_before_expansion':

```
{'rotor_delay_seconds': 0.2, 'gimbal_delay_seconds': 0.040217049712, 'cost':  
1.4156315076964658}, 'added': [{'axis': 'gimbal', 'delay_seconds': 0.045215733641}]}
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selected rotor lag 0.2 s (40.0182 publish periods)  
selected gimbal lag 0.040217049712 s (8.04553 publish periods)  
selected objective cost 1.41563116246
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